Mission 3: Triage

The story

"08:10 in Fier. Arben, 67, cannot find his words. The right side of his face hangs and he cannot lift his right arm. His wife last saw him well at 07:30. The ambulance crew opens Synapse and creates his case. In a few seconds they need to know three things: how bad is it, which pipe is probably blocked, and how much time is left to treat him."

Triage is the word for sorting patients by how urgent they are.

What you need to know

A stroke patient can get two treatments. Both only work for a limited time after the patient was **last seen well**: the last moment someone saw them normal.

1. **The drug** dissolves the clot. It can be given up to **4 hours 30 minutes** after last seen well.
2. **Clot removal**: a doctor pulls the clot out with a thin tool. It only fits in a **big** pipe, and it can be done up to **24 hours** after last seen well.

If a patient wakes up with symptoms, nobody knows when the stroke started. Last seen well is then the time they went to sleep, often the day before.

All times in Synapse belong to one day, called *today*. Last seen well can be *yesterday*, and a time limit can end *tomorrow*. We ask for the time now instead of reading the computer clock, so the judges can repeat the same test.

The base challenge

Your goal: from what the ambulance crew sees, Synapse says how bad the stroke is, where it probably is, and how much time is left to treat it. Every case is saved.

Goal 1: Read the symptom list

We give you `data/symptoms.txt`. It uses the same format as the atlas. Load it when the program starts, like the atlas.

```
# SYMPTOM | code | points | what the ambulance crew sees

SYMPTOM | cannot-speak       | 2 | Cannot get words out, or the words are wrong
SYMPTOM | cannot-understand  | 2 | Does not understand simple questions
SYMPTOM | right-arm-weak     | 3 | Cannot hold the right arm up for 10 seconds
SYMPTOM | left-arm-weak      | 3 | Cannot hold the left arm up for 10 seconds
SYMPTOM | right-face-droop   | 1 | The right side of the face hangs when smiling
SYMPTOM | left-face-droop    | 1 | The left side of the face hangs when smiling
SYMPTOM | right-leg-weak     | 3 | Cannot hold the right leg up for 5 seconds
SYMPTOM | left-leg-weak      | 3 | Cannot hold the left leg up for 5 seconds
SYMPTOM | blind-on-right     | 2 | Does not see things on the right side
SYMPTOM | blind-on-left      | 2 | Does not see things on the left side
SYMPTOM | ignores-left-side  | 2 | Ignores people and things on the left side
SYMPTOM | forgets-new-things | 1 | Cannot repeat 3 words after one minute
SYMPTOM | drowsy             | 3 | Hard to keep awake
SYMPTOM | falls-over         | 2 | Cannot stand or walk straight
```

`triage symptoms` prints the list so the crew can look up a code.

```
synapse> triage symptoms
CODE                 POINTS  WHAT THE CREW SEES
cannot-speak         2       Cannot get words out, or the words are wrong
cannot-understand    2       Does not understand simple questions
...
14 symptoms
```

Goal 2: Create a case

`triage new` asks for the patient's details one by one, checks every answer, and creates a case.

- Name: not empty.
- Age: a whole number from 0 to 120.
- City: not empty. Mission 4 checks it against the map.
- Time now and last seen well: `HH:MM`, 24-hour clock.
- Was last seen well yesterday: `yes` or `no`.
- Symptoms: one or more codes from `symptoms.txt`, separated by commas. A code typed twice counts once.

Each case gets an id: `P-` followed by four digits, starting at `P-0001` and counting up. An id is never reused.

Goal 3: Score the patient and compute the time limits

Apply these rules exactly:

1. **Severity** = the sum of the points of the patient's symptoms.
2. **Band**: 1 to 4 is *minor*, 5 to 9 is *moderate*, 10 or more is *severe*.
3. **Drug possible until** = last seen well + 4 h 30 min. The drug is possible if the time now is not later than that.
4. **Clot removal possible until** = last seen well + 24 h. Clot removal is possible if the time now is not later than that, the likely blocked pipe (Goal 4) is **big**, and the severity is 5 or more.



Goal 4: Find the likely blocked pipe

The crew sees symptoms, not pipes. Synapse has to work backwards, from what the patient shows to the pipe that is probably blocked.

- The likely blocked pipe is a pipe whose blockage would cause **exactly** the patient's symptoms, in any order.
- If more than one pipe matches, pick the one that leaves the fewest pipes without blood. Still tied: the one listed first in the atlas.
- If no pipe matches exactly, the blocked pipe is *unknown*, and clot removal is not possible.

Example CLI interaction (you may design yours differently):

```
synapse> triage new
Patient name: Arben Hoxha
Age: 67
City: Fier
Time now (HH:MM): 08:10
Last seen well (HH:MM): 07:30
Was that yesterday? (yes/no): no
Symptoms (comma separated): cannot-speak, cannot-understand, right-arm-weak, right-face-droop
SUCCESS: Case P-0001 created and saved.
  Severity:            8 (moderate)
  Likely blocked pipe: left-middle (big)
  Drug possible:       yes, until 12:00 (3 h 50 min left)
  Clot removal:        yes, until 07:30 tomorrow

synapse> triage new
Patient name: Drita Marku
Age: 72
City: Shkoder
Time now (HH:MM): 06:50
Last seen well (HH:MM): 22:30
Was that yesterday? (yes/no): yes
Symptoms (comma separated): left-arm-weak, left-face-droop, left-leg-weak, ignores-left-side
SUCCESS: Case P-0002 created and saved.
  Severity:            9 (moderate)
  Likely blocked pipe: right-neck (big)
  Drug possible:       no, the time limit passed at 03:00
  Clot removal:        yes, until 22:30

synapse> triage new
Patient name: Besnik Kola
Age: 58
City: Kukes
Time now (HH:MM): 10:00
Last seen well (HH:MM): 09:00
Was that yesterday? (yes/no): no
Symptoms (comma separated): left-arm-weak, ignores-left-side
SUCCESS: Case P-0003 created and saved.
  Severity:            5 (moderate)
  Likely blocked pipe: unknown, no single pipe matches these symptoms exactly
  Drug possible:       yes, until 13:30 (3 h 30 min left)
  Clot removal:        no, the blocked pipe is unknown
```

Goal 5: Save and show cases

Save every case to your database file, for example `data/cases.json`. Cases must still be there after the program is closed and opened again.



- `triage list` lists every case with its id, name, age, city, severity and likely pipe.
- `triage show <id>` shows one case in full.

Example database structure (you may design yours differently):

```
{
  "cases": [
    {
      "id": "P-0001",
      "name": "Arben Hoxha",
      "age": 67,
      "city": "Fier",
      "createdAt": "08:10",
      "lastSeenWell": "07:30",
      "lastSeenWellYesterday": false,
      "symptoms": ["cannot-speak", "cannot-understand", "right-arm-weak", "right-face-droop"],
      "severity": 8,
      "band": "moderate",
      "likelyPipe": "left-middle",
      "bigPipe": true,
      "drugUntil": "12:00",
      "drugPossible": true,
      "clotUntil": "07:30 tomorrow",
      "clotPossible": true
    }
  ]
}
```

```
synapse> triage list
ID       NAME          AGE  CITY     SEVERITY       LIKELY PIPE
P-0001   Arben Hoxha   67   Fier     8 moderate     left-middle
P-0002   Drita Marku   72   Shkoder  9 moderate     right-neck
P-0003   Besnik Kola   58   Kukes    5 moderate     unknown

synapse> triage show P-0002
=== Case P-0002: Drita Marku ===
Age:                 72
City:                Shkoder
Created at:          06:50
Last seen well:      22:30 yesterday
Symptoms:            left-arm-weak, left-face-droop, left-leg-weak, ignores-left-side
Severity:            9 (moderate)
Likely blocked pipe: right-neck (big)
Drug possible:       no, the time limit passed at 03:00
Clot removal:        yes, until 22:30
```

Error handling

Your program must handle and clearly report errors such as:

- "ERROR: Name cannot be empty."
- "ERROR: Age must be a whole number from 0 to 120."
- "ERROR: Invalid time "8:1O". Use HH:MM, for example 08:10."
- "ERROR: Last seen well (09:30) is after the time now (08:10). Was it yesterday?"
- "ERROR: Unknown symptom "cant-speak". Type triage symptoms to see the list."
- "ERROR: Enter at least one symptom."



- "ERROR: No case called "P-0099"."
- "ERROR: Could not save the case to data/cases.json."

After an error, ask the same question again instead of starting over.

The bonus challenge

For extra points.

Bonus 1: Closest match

Real patients rarely match a pipe exactly. When no pipe matches exactly, score every pipe by similarity:

> *similarity = symptoms in both lists ÷ symptoms in either list*

Show the top 3 as whole percentages, best first. Ties go to the pipe that leaves the fewest pipes without blood. The likely blocked pipe is the best one if it scores **60% or more**; otherwise it stays unknown.

```
synapse> triage new
...
Symptoms (comma separated): left-arm-weak, ignores-left-side
SUCCESS: Case P-0003 created and saved.
  Severity:            5 (moderate)
  No exact match. Closest pipes:
    right-middle          67%
    right-middle-bottom   50%
    right-neck            50%
  Likely blocked pipe: right-middle (big), 67% match
  Drug possible:       yes, until 13:30 (3 h 30 min left)
  Clot removal:        yes, until 09:00 tomorrow
```

Bonus 2: The patient gets worse

Symptoms can change on the way to the hospital. `triage update <id>` asks for the time now and the new symptoms, adds them to the case, and computes everything again. The case keeps a history of every update.

```
synapse> triage update P-0001
Time now (HH:MM): 08:40
New symptoms (comma separated): right-leg-weak
SUCCESS: Case P-0001 updated.
  Severity:            11 (severe), was 8
  Likely blocked pipe: left-neck (big), was left-middle
  Drug possible:       yes, until 12:00 (3 h 20 min left)
```

Bonus 3: Most urgent first

`triage urgent` asks for the time now and lists the cases that can still be treated, the shortest time left first.



```
synapse> triage urgent
Time now (HH:MM): 10:30
Drug still possible:
  P-0001  Arben Hoxha   until 12:00   1 h 30 min left
  P-0003  Besnik Kola   until 13:30   3 h 00 min left
Only clot removal still possible:
  P-0002  Drita Marku   until 22:30   12 h 00 min left
```

Hints

- Time is the trap in this mission. Before you write any code, decide how your program compares 23:50 yesterday with 00:20 today, and how it adds 24 hours to a time.
- The crew types symptoms, but the atlas talks about pipes. You already built something that connects the two.
- The judges load a different atlas and a different symptom list. Anything your program knows that is not in a file will give wrong answers.

What the judges will test

- The three example patients, then hidden ones.
- Midnight: last seen well 23:50 yesterday, time now 00:20.
- The exact limit: a time now equal to the drug limit still counts as possible.
- Closing and reopening the program: every case is still there.
- A hidden `symptoms-v2.txt` with changed points, and `atlas-v2.txt`.

What this mission saves

One case per patient in your database. Mission 4 reads the city, the time now and the two time limits. Mission 7 prints everything.